

11.9 Representations of Functions as Power Series

- Recall that

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n \quad |x| < 1$$

- Example** Express $\frac{1}{1+x^2}$ as the sum of a power series and find the interval of convergence.

$$\begin{aligned} \frac{1}{1+x^2} &= \frac{1}{1-(-x^2)} \\ &= 1 + (-x^2) + (-x^2)^2 + (-x^2)^3 + \cdots \\ &= 1 - x^2 + x^4 - x^6 + \cdots \\ &= \sum_{n=0}^{\infty} (-1)^n x^{2n} \end{aligned}$$

$$\text{interval of convergence} : |-x^2| < 1$$

$$x^2 < 1$$

$$-1 < x < 1$$

$$|x| < 1$$

$$\text{radius of convergence} : R = 1$$

- **Example** Find a power series representation for $\frac{1}{x+2}$.

$$\frac{1}{x+2} = \frac{1}{2} \left[\frac{1}{\frac{x}{2} + 1} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1 - (-\frac{x}{2})} \right]$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2} \right)^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{1}{2^{n+1}} x^n$$

interval of convergence :

$$\left| -\frac{x}{2} \right| < 1$$

$$\frac{|x|}{2} < 1$$

$$|x| < 2$$

radius of convergence :

$$R = 2$$

- **Example** Find a power series representation for $\frac{x^3}{x+2}$.

$$\frac{x^3}{x+2} = x^3 \left(\frac{1}{x+2} \right)$$

$$= x^3 \sum_{n=0}^{\infty} (-1)^n \frac{1}{2^{n+1}} x^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{1}{2^{n+1}} x^{n+3}$$

interval of convergence :

$$|x| < 2$$

radius of convergence :

$$R = 2$$

- **Theorem (term-by-term differentiation and integration)**

If the power series $\sum c_n(x-a)^n$ has radius of convergence $R > 0$, then the function f defined by

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \dots$$

is differentiable on the interval $(a-R, a+R)$ and

$$(i) \quad f'(x) = \sum_{n=1}^{\infty} n c_n(x-a)^{n-1} = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + 4c_4(x-a)^3 + \dots$$

$$(ii) \quad \int_a^x f(t) dt = \sum_{n=0}^{\infty} \frac{c_n(x-a)^{n+1}}{n+1} = c_0(x-a) + \frac{1}{2}c_1(x-a)^2 + \frac{1}{3}c_2(x-a)^3 + \dots$$

The radii of convergence of the power series in (i) and (ii) are both R .

- **Example** Determine the radius of convergence for

$$1 + 2x + 3x^2 + 4x^3 + \dots$$

and find an explicit formula for this series when it is convergent.

$$1 + 2x + 3x^2 + 4x^3 + \dots = \frac{d}{dx} (x + x^2 + x^3 + x^4 + \dots)$$

convergent when $|x| < 1$

$$= \frac{d}{dx} (1 + x + x^2 + x^3 + x^4 + \dots)$$

but this may not be the interval of convergence !

$$= \frac{d}{dx} \left(\frac{1}{1-x} \right)$$

in this example, we can use test for divergence to show that the series diverges when $x = \pm 1$. Therefore, interval of convergence is $|x| < 1$.

$$= \frac{d}{dx} (1-x)^{-1}$$

$$= (-1)(1-x)^{-2}(-1)$$

$$= \frac{1}{(1-x)^2}$$

Radius of convergence :
 $R = 1$

- **Example** Determine the interval of convergence for

$$x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots$$

and find an explicit formula for this series when it is convergent.

$$x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots$$

$$= \int_0^x (1 + t + t^2 + t^3 + \dots) dt$$

$$= \int_0^x \frac{1}{1-t} dt$$

$$= -\ln |1-t| \Big|_0^x$$

$$= -\ln |1-x|, \quad |x| < 1$$

$$\underline{x=1} : 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

This series diverges as it is the harmonic series.

$$\underline{x=-1} : -1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \dots$$

This is the negative of alternating harmonic series.

Since the alternating harmonic series converges, the negative version converges as well.

$$\text{Interval of convergence} : [-1, 1)$$

$$\text{Radius of convergence} : R = 1$$

- **Example** Find the power series representation for $\ln(1+x)$ and determine its radius of convergence.

$$\ln(1+x) = \int_0^x \frac{1}{1+t} dt$$

$$= \int_0^x \frac{1}{1-(-t)} dt$$

$$= \int_0^x (1 - t + t^2 - t^3 + \dots) dt$$

$$= t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \dots \Big|_0^x$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \quad |x| < 1$$

↑

interval of convergence
is $(-1, 1]$.

Radius of convergence : $R = 1$.

- **Example** Find the power series representation for $\tan^{-1} x$.

$$\begin{aligned}
 \tan^{-1} x &= \int_0^x \frac{1}{1+t^2} dt \\
 &= \int_0^x \frac{1}{1-(-t^2)} dt \\
 &= \int_0^x \sum_{n=0}^{\infty} (-t^2)^n dt \\
 &= \sum_{n=0}^{\infty} (-1)^n \left(\int_0^x t^{2n} dt \right) \\
 &= \sum_{n=0}^{\infty} (-1)^n \left[\frac{t^{2n+1}}{2n+1} \right]_{t=0}^{t=x} \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}, \quad |x| < 1
 \end{aligned}$$

radius of convergence : $R = 1$.

interval of convergence : $[-1, 1]$.

- **Example** Find a formula for the sum of the series

$$S(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

Note that $S(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges everywhere from P. 52.

$$S'(x) = 0 + 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = S(x)$$

$$\text{Let } y = S(x).$$

$$\frac{dy}{dx} = y$$

$$\int \frac{dy}{y} = \int dx$$

$$\ln |y| = x + C$$

$$|y| = e^{x+C} \Rightarrow y = Ae^x$$

$$\text{When } x = 0, y = S(0) = 1 :$$

$$1 = Ae^0 \Rightarrow A = 1$$

$$\therefore y = e^x$$

$$\therefore S(x) = e^x, \quad x \in \mathbb{R}$$

• **Example**

(a) Evaluate $\int \frac{1}{1+x^7} dx$ as a power series.

$$\begin{aligned}
 \int \frac{1}{1+x^7} dx &= \int \frac{1}{1-(-x^7)} dx \\
 &= \int \sum_{n=0}^{\infty} (-x^7)^n dx \\
 &= \sum_{n=0}^{\infty} (-1)^n \int x^{7n} dx \\
 &= C + \sum_{n=0}^{\infty} (-1)^n \frac{x^{7n+1}}{7n+1}
 \end{aligned}$$

interval of convergence: $(-1, 1]$

converges when $|-x^7| < 1 \Leftrightarrow |x| < 1$

radius of convergence: $R = 1$

(b) Approximate $\int_0^{\frac{1}{2}} \frac{1}{1+x^7} dx$ correct to within 10^{-7} .

$$\int_0^{\frac{1}{2}} \frac{1}{1+x^7} dx = \left[\sum_{n=0}^{\infty} (-1)^n \frac{x^{7n+1}}{7n+1} \right]_0^{\frac{1}{2}} = \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{1}{2}\right)^{7n+1}}{7n+1}$$

Alternating series: $|S - S_n| \leq b_{n+1} < 10^{-7}$

$$\times \quad b_n = \frac{\left(\frac{1}{2}\right)^{7n+1}}{7n+1}$$

$$n=2 : b_3 = \frac{\left(\frac{1}{2}\right)^{23}}{23} = 1.08 \times 10^{-8}$$

$$\therefore \int_0^{\frac{1}{2}} \frac{1}{1+x^7} dx = \underbrace{(-1)^0 \frac{\left(\frac{1}{2}\right)^1}{1}}_{n=0} + \underbrace{(-1)^1 \frac{\left(\frac{1}{2}\right)^8}{8}}_{n=1} + \underbrace{(-1)^2 \frac{\left(\frac{1}{2}\right)^{15}}{15}}_{n=2} = 0.4995138$$

11.10 Taylor and Maclaurin Series

• Theorem

If f has a power series representation (expansion) at a :

$$f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots \quad \text{when } |x-a| < R,$$

then

$$c_n = \frac{f^{(n)}(a)}{n!}.$$

$$f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots$$

$$f(a) = c_0 + c_1(a-a) + c_2(a-a)^2 + c_3(a-a)^3 + \cdots = c_0$$

$$f'(x) = 0 + c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \cdots$$

$$f'(a) = c_1$$

$$f''(x) = 0 + 0 + 2c_2 + 6c_3(x-a) + \cdots$$

$$f''(a) = 2c_2$$

$$f'''(x) = 0 + 0 + 0 + 6c_3 + \cdots$$

$$f'''(a) = 6c_3$$

The pattern shows $c_n = \frac{f^{(n)}(a)}{n!}$

- Substituting $c_n = \frac{f^{(n)}(a)}{n!}$ back into the series, we obtain

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \\ &= f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \frac{f'''(a)}{3!} (x-a)^3 + \dots \end{aligned}$$

- The above series is called the **Taylor series** for the function f at a .
- If $a = 0$, the series is called the **Maclaurin series**.
- **Example** Find the Maclaurin series of the function $f(x) = e^x$ and determine its radius of convergence.

$$f(x) = e^x$$

$$f^{(n)}(0) = e^0 = 1$$

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

From 1.52, the series converges everywhere.

$$\therefore R = \infty.$$

- n^{th} -degree Taylor polynomial of f at a : $T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$
- $f(x)$ is the sum of its Taylor series if $f(x) = \lim_{n \rightarrow \infty} T_n(x)$.
- **Remainder** of the Taylor series: $R_n(x) = f(x) - T_n(x)$

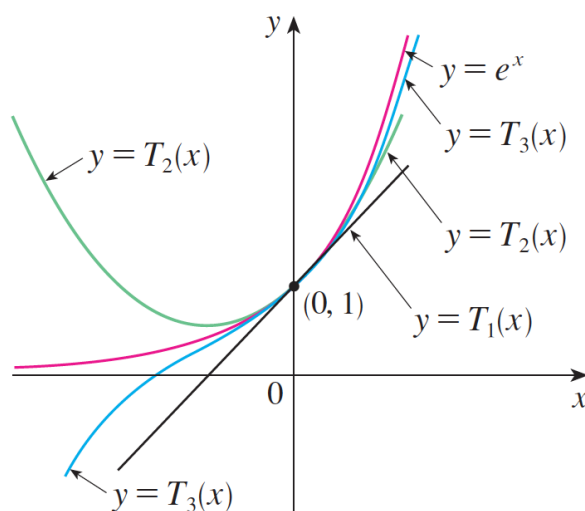
$$f(x) = e^x$$

$$T_n(x) = \sum_{k=0}^n \frac{x^k}{k!}$$

$$T_1 = 1 + x$$

$$T_2 = 1 + x + \frac{x^2}{2!}$$

$$T_3 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$$



• Theorem

If $f(x) = T_n(x) + R_n(x)$ and

$$\lim_{n \rightarrow \infty} R_n(x) = 0$$

for $|x - a| < R$, then f is equal to the sum of its Taylor series on the interval

$$|x - a| < R$$

$$\lim_{n \rightarrow \infty} R_n(x) = 0 \Rightarrow \lim_{n \rightarrow \infty} [f(x) - T_n(x)] = 0$$

$$\Rightarrow f(x) = \lim_{n \rightarrow \infty} T_n(x)$$

- **Taylor's Inequality**

If $|f^{(n+1)}(x)| \leq M$ for $|x - a| \leq d$, then the remainder $R_n(x)$ of the Taylor series satisfies the inequality

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1} \quad \text{for } |x - a| \leq d$$

- In showing $\lim_{n \rightarrow \infty} R_n = 0$, it is often helpful to make use of the following fact.

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad \text{for every real number } x$$

From P-52, $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges everywhere.

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad \text{for all } x$$

- **Example** Prove that e^x is equal to the sum of its Maclaurin series.

$$f(x) = e^x$$

$$f^{(n+1)}(x) = e^x$$

$$\text{For } |x| \leq d, \quad |f^{(n+1)}(x)| \leq e^d$$

$$\text{so, } |R_n(x)| \leq \frac{e^d}{(n+1)!} |x|^{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{e^d}{(n+1)!} |x|^{n+1} = e^d \left(\lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{(n+1)!} \right) = 0$$

$\downarrow$
 0

$$\therefore \lim_{n \rightarrow \infty} R_n(x) = 0 \quad \text{by Squeeze Theorem}$$

$\therefore e^x$ is equal to the sum of its Maclaurin series.

- **Example** Find the Taylor series for e^x at a and determine its radius of convergence.

$$f(x) = e^x$$

$$f^{(n)}(x) = e^x \quad \Rightarrow \quad f^{(n)}(a) = e^a$$

$$f(x) = \sum_{n=0}^{\infty} \frac{e^a}{n!} (x-a)^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{e^a}{(n+1)!} (x-a)^{n+1}}{\frac{e^a}{n!} (x-a)^n} \right| < 1$$

0
||

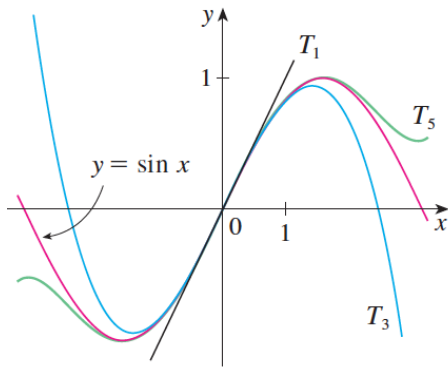
$$\left(\lim_{n \rightarrow \infty} \frac{1}{n+1} \right) |x-a| < 1$$

$$-\infty < x-a < \infty$$

$$-\infty < x < \infty$$

$\therefore$ Radius of convergence : $R = \infty$.

- **Example** Find the Maclaurin series for $\sin x$ and prove that it represents $\sin x$ for all x .



$$f(x) = \sin x$$

$$f(0) = 0$$

$$f'(x) = \cos x$$

$$f'(0) = 1$$

$$f''(x) = -\sin x$$

$$f''(0) = 0$$

$$f'''(x) = -\cos x$$

$$f'''(0) = -1$$

$$f^{(4)}(x) = \sin x$$

$$f^{(4)}(0) = 0$$

$$\vdots$$

$$\vdots$$

$$\sin x = 0 + \frac{x}{1!} + 0 - \frac{x^3}{3!} + 0 + \dots$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$|f^{(n+1)}(x)| \leq 1 \quad \text{for all } x.$$

$$\text{so, } |R_n(x)| \leq \frac{1}{(n+1)!} |x|^{n+1} \xrightarrow{n \rightarrow \infty} 0$$

$\therefore \sin x$ is equal to the sum of its Maclaurin series.

It can be shown that $R = \infty$.

- **Example** Find the Maclaurin series for $\cos x$.

$$\begin{aligned}
 \cos x &= \frac{d}{dx} \sin x \\
 &= \frac{d}{dx} \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)!} \left(\frac{d}{dx} x^{2n+1} \right) \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)!} \left((2n+1) x^{2n} \right) \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}
 \end{aligned}$$

radius of convergence:
 $R = \infty$

- **Example** Find the Maclaurin series for $x \cos x$.

$$\begin{aligned}
 x \cos x &= x \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n)!}
 \end{aligned}$$

radius of convergence:
 $R = \infty$

- **Example Binomial Series** Find the Maclaurin series for $(1+x)^k$ and show that it is convergent for $|x| < 1$.

$$f(x) = (1+x)^k$$

$$f(0) = 1$$

$$f'(x) = k(1+x)^{k-1}$$

$$f'(0) = k$$

$$f''(x) = k(k-1)(1+x)^{k-2}$$

$$f''(0) = k(k-1)$$

$$f'''(x) = k(k-1)(k-2)(1+x)^{k-3}$$

$$f'''(0) = k(k-1)(k-2)$$

$\vdots$

$$(1+x)^k = 1 + kx + \underbrace{\frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \dots}_{\frac{k(k-1)\dots(k-n+1)}{n!}x^n} + \dots$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{k(k-1)\dots(k-n+1)(k-n)}{(n+1)!}x^{n+1}}{\frac{k(k-1)\dots(k-n+1)}{n!}x^n} \right| < 1$$

$$\left(\lim_{n \rightarrow \infty} \frac{|k-n|}{n+1} \right) |x| < 1$$

$$|x| < 1 \Rightarrow \text{radius of convergence : } R = 1$$

* if $-1 < k \leq 0$, interval of convergence : $(-1, 1]$.

* if $k \geq 0$, interval of convergence : $[-1, 1]$.

- **Example** Find the Maclaurin series of $\frac{1}{(1-x)^2}$ and determine its radius of convergence.

$$\begin{aligned}
 \frac{1}{(1-x)^2} &= [1 + (-x)]^{-2} \\
 &= 1 + (-2)(-x) + \frac{(-2)(-3)}{2!} (-x)^2 \\
 &\quad + \frac{(-2)(-3)(-4)}{3!} (-x)^3 + \dots \\
 &= 1 + 2x + 3x^2 + 4x^3 + \dots
 \end{aligned}$$

$$|-x| < 1 \quad \Leftrightarrow \quad |x| < 1$$

$\therefore$ radius of convergence : $R = 1$.

- **Example** Find the first 4 nonzero terms in the Maclaurin series of $\frac{1}{\sqrt{4-x}}$ and determine its radius of convergence.

$$\begin{aligned}
 \frac{1}{\sqrt{4-x}} &= \frac{1}{2} \left[1 + \left(-\frac{x}{4}\right) \right]^{-\frac{1}{2}} \\
 &= \frac{1}{2} \left[1 + \left(-\frac{1}{2}\right)\left(-\frac{x}{4}\right) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!} \left(-\frac{x}{4}\right)^2 \right. \\
 &\quad \left. + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!} \left(-\frac{x}{4}\right)^3 + \dots \right] \\
 &= \frac{1}{2} \left(1 + \frac{x}{8} + \frac{3x^2}{128} + \frac{5x^3}{1024} + \dots \right) \\
 &= \frac{1}{2} + \frac{x}{16} + \frac{3x^2}{256} + \frac{5x^3}{2048} + \dots
 \end{aligned}$$

$$\left| -\frac{x}{4} \right| < 1 \quad \Leftrightarrow \quad |x| < 4$$

$\therefore$ radius of convergence : $R = 4$.

- Important Maclaurin Series

1. $\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots \quad |x| < 1 \quad \sum_{n=0}^{\infty} x^n$
2. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots \quad -1 < x \leq 1 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$
3. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \dots \quad -1 \leq x \leq 1 \quad \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$
4. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \quad x \in \mathbb{R} \quad \sum_{n=0}^{\infty} \frac{x^n}{n!}$
5. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots \quad x \in \mathbb{R} \quad \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$
6. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots \quad x \in \mathbb{R} \quad \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$
7. $(1+x)^k = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \dots \quad |x| < 1$

- Example** Find the sum of the series $\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} - \frac{1}{4 \cdot 2^4} + \dots$

$$\begin{aligned}
 \frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} - \frac{1}{4 \cdot 2^4} + \dots &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n \cdot 2^n} \\
 &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\left(\frac{1}{2}\right)^n}{n} \\
 &= \ln\left(1 + \frac{1}{2}\right) \\
 &= \ln \frac{3}{2}
 \end{aligned}$$

• **Example**

(a) Evaluate $\int e^{-x^2} dx$ as an infinite series.

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{n!}$$

$$\begin{aligned} \int e^{-x^2} dx &= \int \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{n!} dx \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \left(\int x^{2n} dx \right) \\ &= C + \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \frac{x^{2n+1}}{2n+1} \end{aligned}$$

(b) Evaluate $\int_0^1 e^{-x^2} dx$ correct to within an error of 0.001.

$$\int_0^1 e^{-x^2} dx = \left[\sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \frac{x^{2n+1}}{2n+1} \right]_{x=0}^{x=1} = \sum_{n=0}^{\infty} (-1)^n \frac{1}{n! (2n+1)}$$

b_n
" 1

Alternating series : $|S - S_n| \leq b_{n+1} < 0.001$

$$n=3 : b_4 = \frac{1}{4!(9)} = 0.0046$$

$$n=4 : b_5 = \frac{1}{5!(11)} = 0.00076 \quad (\checkmark)$$

$$\begin{aligned} \int_0^1 e^{-x^2} dx &= \frac{1}{0!(1)} - \frac{1}{1!(3)} + \frac{1}{2!(5)} - \frac{1}{3!(7)} + \frac{1}{4!(9)} \\ &= 0.7475 \end{aligned}$$